AaS! 283: What's Really Going on with UAPs?

Podcast: Ask a spaceman

AI Executive Summary

1. Core Thesis This episode argues that UFOs/UAPs should be approached not as a binary of believers vs. debunkers, but as an open scientific inquiry into unidentified phenomena where "unidentified" is the most crucial fact, not the mystery itself. The speaker posits that while human meaning-making around mysteries is valid, and extraterrestrial hypotheses are theoretically possible, professional science favors terrestrial or natural explanations because they require fewer evidentiary leaps to be plausible. Without high-quality, repeatable data, the burden of proof lies heavily on those claiming alien identity, as ruling *out* other possibilities is scientifically rigorous without confirming one over the others.

2. Key Arguments & Data

3. **Definition and Terminology:** The term UAP (Unidentified Aerial Phenomena) replaces UFO to include non-flying objects (floats, falls); however, "unidentified" remains the core definition, meaning we have no confirmed knowledge of what it is rather than assuming a specific identity.
4. **Evidence Standards:** It is easier to claim something is unidentified but significantly harder to make the positive claim that it *is* extraterrestrial. Ruling in requires satisfying high evidentiary standards (existence of advanced interstellar travel, interest in Earth), whereas ruling out relies on existing knowledge and plausibility.
5. **Political & Safety Factors:** Congressional hearings legitimize reports but primarily address safety/security concerns for military pilots; sworn testimony is not the same as evidence that can be examined independently by others.
6. **Hypothesis Spectrum (Ranked by Plausibility):** In data-poor environments, the scientific method involves ranking hypotheses. The most common and scientifically parsimonious explanations are:
    - Human-made technology (drones, satellites, secret government programs, balloons).

    - Natural phenomena (optical illusions like lenticular clouds, mirages, sun dogs, ball lightning, Venus at low horizon angles).
    - Instrument/Sensor Glitches (miscalibration, lens artifacts, misinterpretation of raw data).
7. **Human Cognitive Biases:** Humans are exceptionally bad at judging distance, motion, and acceleration in the sky. Expectations bias pattern recognition (e.g., seeing a spaceship shape where there is none), and eyewitness accounts change with retelling due to memory malleability.
8. **Case Study:** A Navy video of an object moving at high speed was analyzed by the NASA UAP panel; once sensor motion was subtracted, the object was determined to be piece of junk drifting in the wind at wind speeds, proving the object was not defying physics but merely misidentified due to poor calibration.

9. **Data Quality:** Professional astronomical interest is low in this sector because observations rely mostly on rough, poor-quality eyewitness accounts or low-quality recordings compared to other science fields that have richer data for analysis.

10. Actionable Takeaways

11. **Apply Rigorous Skepticism:** Treat all observation extensions (eyewitnesses and sensors alike) with suspicion. Do not accept an object as "alien" based on a single report without further evidence or triangulation.
12. **Evaluate Simplicity:** When evaluating unidentified sightings, default to the simplest explanation unless data proves otherwise. Consider that human-made debris or optical effects are far more probable than alien technology.
13. **Advocate for Better Data:** If interested in UAPs/UFOs, support initiatives that improve data quality (better instrumentation and sensor calibration) rather than relying on anecdotal accounts to make progress.
14. **Respect Meaning Making:** Acknowledge the human value of these mysteries; finding meaning or community around unidentified phenomena is a valid aspect of the experience, even if it doesn't align with strict scientific proof.

15. **Understand "Unidentified":** Recognize that acknowledging uncertainty ("we don't know what this is") is scientifically accurate and does not require belief in a specific entity to remain honest about the state of evidence.

Full Transcript

Wait, this is an episode about aliens? Yes, it's an episode about aliens, but not in the way you might think. It's really an episode about scientific thinking and skepticism and the many mysteries of the world that surround us. But we're going to use aliens, UFOs, UAPs, and other assorted related ideas to talk about that stuff. And I'm doing it now because UAPs or UFOs will get to this in a second.

I keep popping up in the news and in congressional hearings and listeners keep asking about it and I want to set expectations. This is not a debunking episode and it's not a believer episode either. The most important word when it comes to unidentified flying objects is that word unidentified and what we do about it. First I need to clear the air. So to speak, we need to define what a UFO or UAP is. Now, when I was growing up, it was straight up 100% always UFO, unidentified flying object. These are the saucers, the beams, the things doing things to cows, doing things to people, the great aliens with the big eyes visiting in their spaceships and observing, probing, experimenting, doing experiments of questionable scientific merit and ethical scrutiny on humanity. In recent years, UFOs have gotten a bit of a rebrand. Part of that is valid. And I appreciate it. And part of it as well rebranding. UFOs became not cool. They became something to make fun of. They became unserious. I mean, even less serious than they were. And they didn't exactly have far to go, but they became an object of ridicule rather than one of potential curiosity and scrutiny.

So when serious people wanted to start talking about UFOs, they would get laughed out of the room and nobody likes getting laughed out of the room. Even comedians, they prefer to get laughed into the room anyway. Hence the rebranding to UAP or Unidentified Aerial Phenomena. Now there is a validity to this because not everything in the air is flying, so it covers a broader class of object. Things can float, things can fall. Okay, those were the only two other options I could think of. But hey, that's two other options other than fly. And in aerial just means stuff happening in the air. And so this is this helps an object implies a thing. It could be a vehicle, it could be a spaceship, it could be a device. Phenomena is a broader class of entity of

things that can anything that can happen up in the sky that we can't fully explain.

So, UAP, okay, I'll take it. I also understand it's mostly just about rebranding, but I can also take it. But in either case, the keyword is unidentified, and that's going to be central to our discussion here. Whether it's an unidentified flying object or an unidentified aerial phenomenon, what matters is that it's unidentified. And unidentified means we don't know what it is. I'm going to tell you the punchline because I like doing that in my episodes. I'll tell you the punchline up front and then we'll work backwards to explain it. And the punchline is this. Unidentified means we don't know what it is. That's as far as the evidence allows in some cases. But the moment you say you know what it is, if you say it's aliens, well, that's a different kind of claim. It's easy to say something is unidentified. It's a lot harder to say you know what it is.

these require different levels of evidence and they require different standards. And this is my general response whenever someone tells me a UFO story. You may not be surprised by I hear a lot of UFO stories, I appear on radio shows and podcasts, I have conversations with fans at events, and maybe some of you have had an experience with that UAP. And I'll be honest with you, so have I. There's been stuff in the sky that I've seen that I can't identify. In fact, I would say just about all of us have at one point in our lives seen something in the sky thought, huh, what the heck is that? And then moved on. And in the strictest or least strict, depending on how you feel today, definition of UAP, we have all encountered an unidentified aerial phenomenon. So when someone tells me a UFO or UAP story, my first instinct is to listen.

We've all got crazy stories, stories about stuff we don't understand. That's like all of science, right? We live in a universe that we are struggling to understand and we're developing some techniques to try to understand it. If the universe wasn't full of mysteries, A, it'd be a lot more boring and B, I'd be out of a job. So I'm quite a fan of mysteries in the universe. For most people, when they encounter something in the sky that they don't fully understand, that's about the long and short of it. They see something in the sky, they go, huh, that's weird. And then they go back to work, back to their

relationship or back to nibbling on that cheese they were enjoying, whatever they were doing. And that event leaves their mind as quickly as it entered. But for some people, this becomes a big deal. And I'm not mocking that. We all have things in our lives that are mundane or boring for nine out of 10 people in nine out of 10 situations. But for that one person out of 10 in that one happenstance circumstance, it becomes meaningful. Maybe it's a song you hear on the radio.

or a glance a stranger gives you or something you find in the attic. We find meaning in this world in a variety of ways. And for some people, seeing something mysterious up in the sky is meaningful. Heck, I look up in the sky and I find meaning there. Good, clear, dark sky. I'm drawn to it. It's meaningful to me. When I do my work of cosmology and I try to understand the evolution of the universe of the voids between the galaxies, that carries meaning to me beyond just a day job. It's meaningful. And so my first job as a scientist and science communicator and as a general human being is to never rob meaning from someone's encounter with the natural world. I derive a lot of meaning from my own work and my own experiences. And who am I to take that away from someone else? And what do people do when they find meaning in a shared experience? They build a community around it. That's not weird.

or unnatural or even surprising. So we have a bunch of people who for various reasons find meaning and significance and importance in unidentified things that show up in our sky and they build communities around it. That's my take on the whole UFO UAP thing, like why there are communities around it because people are drawing meaning from the mysterious. There's another side to that coin, which is when it reaches beyond the levels of a community. When there are things like congressional hearings on the subject, which crop up once or twice a year. And there are a few reasons why I think we see congressional hearings on UFOs, UAPs. Some members of Congress are just interested in it. They do, after all, represent us and plenty of their constituents care about this. There are congressional hearings and meetings on the future of science on big astronomy projects on NASA missions. When people in a representative democracy care about something, then eventually the people who represent them in that democracy will talk about it in their official capacity. There is

also a real safety and security issue of if military pilots keep seeing unknown objects near training ranges or bases. That's a problem. Whether it's aliens, rival country drones, a hobbyist quadricopter, whatever it is. Oh, we got to talk about that. We let's let's figure that out. hearings also legitimize. UAP reports. Remember, We'll get to plausible reasons of what could explain some of these unidentified things. But one potential thing is, you know, an enemy's drones or rival powers, drones, technology that we don't fully understand and getting pilots actually report that has been a problem. This is part of the reason of the rebranding. If you reported a UFO, it could hurt your career. So sightings.

went unrecorded and we kind of like to know what our pilots are seeing, especially around sensitive areas. So the Pentagon's UAP office, which was set up in 2022, are designed to remove that stigma, having congressional hearings are designed to remove that stigma to get our pilots talking about what they see, even who cares if it's aliens, we just want pilots reporting what they see. And also in typical political maneuverings, it's a distraction. A splashy alien hearing is a lot more fun to cover than yet another budget fight. It's ways for members of Congress to poke at secret programs and classified spending that are otherwise hard to get. And yeah, but that's as far as I take the hearings. Yes, they can have dramatic claims. And yeah, the claims are made sworn under oath, but a sworn testimony is not the same thing as evidence that someone else can examine. So I see the purpose.

The multifaceted purpose of congressional hearings around UAPs, UFOs, but that's about all I take from it. At the end of the day, it's stories. And some of these stories are factual reportings of observations and some of the stories are, let's say, have more meaning attached to it. And if it stopped there, I wouldn't have much more to say. The same way I wouldn't necessarily devote an episode to people liking a particular musician or being really fond of the color mauve and building communities around that. That's what we do. The reason I'm doing an episode on it is that UFOs and UIPs naturally intersect science and they do it in two ways. One, they're space and technology adjacent, which is also very science adjacent, especially in the modern era. And two, they're, you know, unidentified. You know why you like your favorite artist. Or maybe you don't know it at a

deep biochemical psychological level, but there's not much to poke at there. You don't know why you like the color mob. You just know that you do.

And you can like or be interested or fascinated by UFOs or UAPs, that's not the question. The question is, if they're unidentified, then what are they? And when something in the natural world is unidentified, we have a tendency to want to identify it. It's kind of a thing that humanity has been doing for a pretty long time and won the latest versions of handling those kinds of questions as science. And so the examination of UAPs and UFOs falls squarely... under the umbrella of science at least at first glance. And because science is concerned with identifying things and understanding objects that appear in our universe, and fields like astronomy are especially designed to understand things that appear in our sky, for various definitions of the word sky, we feel an urge to explain the unexplained. And here's where things get interesting. Because we're dealing with fundamentally unidentified things, we can't assign an identity right off the bat.

Instead, we have a spectrum of ideas. We have a range. We have categories. We have possibilities that we can offer as hypotheses to explain the unexplained. And without further evidence, we are left with the viability and rationality and plausibility of those hypotheses to rank our guesses. Remember, in science, evidence is the ultimate arbiter. But if we don't have a lot of evidence, if we just have a single eyewitness report of something, or a single recording of something, then we play the hypothesis ranking game, knowing that we can't know for sure. So what are some potential hypotheses? Yeah, it's a possibility that an extraterrestrial civilization could have developed a technology for interstellar travel, found the earth, traveled to the earth, and is currently puzzling around the earth either in person or through various insundary robotic emissaries performing experiments on us and or observing us.

It is a possibility. There is nothing in the laws of physics or a current understanding of the universe that makes this outright impossible, so we must hold it open as a hypothesis. But there are other hypotheses. For example, it could be technology, be it mundane technology or advanced technology or even secret technology that is purely 100% terrestrial human made. It could be drones. Could be piloted aircraft. It could be balloons. It

could be stuff that we make and put up in the sky. We've been putting stuff up in the sky in a serious way for a few generations now. We've gotten pretty good at it. And the sky is crowded nowadays. We've got tens of thousands of flights a day. We've got hobby drones. We've got thousands of satellites. A Starlink trains are reported as UFOs all the time. We have weather research balloons going up.

Every day at altitude, they can hang still or drift against the wind. Like you feel the wind blowing in one direction on the ground, but at altitude, the wind is going in a different direction. So it looks odd. We do have governments that do keep secrets from us and do have secret programs with secret technological development, including stuff that flies around in the air doing secret stuff. And then you see it and you're like, oh, my God, that looks weird. What is that? And you call around to your local government officials and they say, I don't know, there's nothing maybe. I don't know. A big chunk of the sightings in the 1950s and 60s where the U2 spy plane flights that at the time were secret, for example, rocket launches at twilight, the exhaust plume lit by the sun produces jellyfish shapes that are visible for hundreds of miles. And if you look at it, you're like, oh, that seems unidentifiable. There it is.

And then there's junk, party balloons, plastic bags, debris, just pollution. There are a lot of us putting stuff up into the sky pretty much all the time. And so we must hold that out as a hypothesis. If we see something unidentified, one hypothesis is that it is something human made that is in the sky. And we will continue with our hypotheses after this commercial break. Welcome back. Another possibility is that it's a natural phenomenon, light bouncing around in a certain way, a meteor or comet. Natural stuff happens in the atmosphere all the time, or out in deep space, and sometimes it's hard to tell which is which. For example, Venus, when it's bright and low on the horizon, is one of the most commonly reported UFOs ever. Fireballs re-entering space junk.

They they move slowly they can break into pieces and it looks nothing like a shooting star There's weird atmospheric optics all the time lenticular clouds. These are the ones that actually look like saucers Sun dogs light pillars. I was up in the Arctic recently and We saw like crazy crazy light

pillars sun dogs. It was just a normal day And you'd look at it and say, what the heck is that? And everyone around me was saying, what the heck is that? I said, oh, that's a sun dog. And they said, well, what the heck is that? And then I had to explain the physics of ice crystals and et cetera, et cetera. Ask me if you want to do an episode on it. Mirages from temperature revision, inversions, light from a distant source gets bent, goes off in weird directions that are unexpected. There's electrical stuff in the atmosphere that's rare, but common enough to be seen.

ball lightning sprites. If you've got a camera, we think we tend to think of cameras and technology as the ultimate like arbiters of truth, like they they record reality as it is. But they're just lenses. They're just digital systems, birds and bugs catching the light right next to a camera lens can look like a distant object. when really it's just something happening right in front of the camera and the camera doesn't know, it's just absorbing photons and recording them in an electrical sensor. And then we look at a picture, a two-dimensional representation of a three-dimensional world, and things can look weird sometimes. It can also just be misunderstanding of our own eyes. The entire apparatus of science is designed to counter the fact that we can easily fool ourselves. We walk into observations with past biases, preconceived notions.

Especially when it comes to pattern matching, we often make a first guess as to what we're seeing and we tend to stick to that guess, especially if what we're seeing is odd or slightly out of context. This is especially true when it comes to the sky. We are fantastically bad at judging distance, motion direction and acceleration in the sky. We're not exactly birds or descended from bird-like creatures. Our reference frame is the ground or maybe trees. When we see something happening on the ground, we are exceptionally good at ballparking distances, movements, relationships between objects. But up in the sky, we lose our reference frame. Something that appears to be close is actually really far away and vice versa. Something that appears to be zigzagging can just be following a normal course. Because we have nothing to compare against a small thing nearby and a huge thing far away look exactly the same. If you stare at a single point of light in a dark sky, it'll actually appear to wander on its own because of the way our eyes work.

in low light conditions. We often have a hard time accounting for our own motion if we're driving or flying or just turning our head. It can make still objects appear to move or to keep pace with you. Our memory is very slippy, malleable. Eyewitness accounts change with every retelling. We think the more we tell a story, the more solid it gets, but research has shown that it's actually the more it changes. Expectations and biases fill in the gaps. If your brain says spaceship once, you didn't even ask your brain. You just saw something weird in the sky and your brain says that's a spaceship. Then all the details start lining up to match spaceship and not something else. That's normal human bias. That's why we have science to filter that out. Of course, not all observations or recordings of UAPs and UFOs rely just on our eyeballs and eyewitness accounts. Sometimes they come from instruments like radar.

Long-range cameras or any variety of military civilian sensors. But this introduces a new class of hypotheses, glitches, malfunctions, bad data misinterpreting the output. If a sensor isn't properly calibrated, it could return readings that look unidentified, but really it's just junky data. If you're looking through a digital camera at a distance source in the middle of the night and your instrument isn't properly calibrated, or isn't accounting for the motion of your own aircraft or sensor gear or whatever, You can get some really anomalous behavior that is difficult to explain because it looks like the object is defying our understanding about aircraft or things in general move through the air. But that's part of the scientific process as well. You don't just distrust your own eyes and ears, you automatically distrust the extensions of your eyes and ears, your telescopes, your sensors, your cameras. In science, we approach all of our observations from a position of skepticism and distrust.

The same rigor we apply to ourselves, we apply to extensions of ourselves, and so a plausible hypothesis for any UAP observation is just bad data. An example of this, one of the famous Navy videos appears to show an object racing over the ocean. When NASA's UAP panel, yes, NASA had a UAP panel, worked through the geometry in 2023, the object was moving at wind speed around 40 miles an hour. The speed, the apparent speed came from the jet's own motion. Once you subtract it out, it was just piece of junk floating in the wind. If your sensor isn't calibrated, it can give consistent

results that are wrong. If your bathroom scale is miscalibrated, you know, it's dependable, it's repeatable, gives you the same weight every single day, sticks to your diet, and it's wrong because it's miscalibrated.

glare, lens artifacts of a distant bright object can bloom into a fuzzy shape that seems to rotate or change, but really it's just the camera mount. It's the lenses, the optics. You can try this yourself. Take a phone camera and photograph a spinning propeller, a fan. You get these bent, weird looking blades that simply don't exist. The camera invents the shape because of the limitations of the physics of capturing a still image of a moving object. And of course, there's like a meta hypothesis to all of this, to bad data, miscalibrated optics, human bias, natural phenomenon, human made artifacts, which is that it's some combination of some or all of the above. A UAP could be a natural phenomena that's observed at the wrong time of day. And that if you saw it an hour later, it would be totally normal and unremarkable, but you saw it then. And you don't have a natural reference frame for it. I mean, Keep in mind, many people are still surprised when they see the moon during the day. This is an example of how preconceived notions can lead us astray. There can be other combinations, an instrument glitch or a failure to account for some sort of sensor, calibration, and noise level that then gets misinterpreted by human bias. And heck, it could be aliens that we see with our own eyes, but in such a confused, obfuscated way that it's not immediately obvious that it's aliens.

literally waving their tentacles at us. And we're like, well, I think it kind of looks like, I don't know, maybe Venus is a little low on the horizon. It's hard. And I'm moving on this boat. It's a little hard to tell. We can have combinations of plausible hypotheses that come together to make something unidentified. But there is something that is identified. There is a strand running through here. And that's Patreon. That's patreon.com slash PM Sutter. No blurry photos, no grainy radar returns. Just you and me. Patreon.com slash PM Sutter. I really appreciate it. Like I said, these are all possibilities. They're all hypotheses that in the absence of data we have to hold on to. And the game of science is to acquire additional data or develop convincing enough arguments to knock out one or more of the hypotheses.

Now, I'm aware that there are several databases and catalogs of UIP incidents and they generally play this game starting with hypotheses and finding the most likely one or at least knocking out one hypothesis or another. Like if you know Venus is not in the sky at the time of the recorded incident, then you know Venus is not responsible for that UIP. And even after this process, there are some recorded UIP incidents and some eyewitness accounts that have no plausible explanation behind them, by which I mean there's no convincing evidence one way or the other about what it is. So we can still rank the remaining surviving hypotheses based on plausibility. We ask ourselves this in science all the time. What is the simplest possible explanation? Even knowing that we can't prove it one way or the other. And this is where aliens become a tough sell. This is where it flips. Saying I don't know is easy saying it's aliens is a lot harder.

You can hold yourself out and say, yeah, it's unidentified and here's some potential hypotheses, but to sink your teeth into one hypothesis, you've got to sell the story, which means the story has to be plausible and fit into everything else we know. Miscalibrated instruments, misunderstood eyewitness observations, natural phenomena, the known fact that the governments of the world have a habit of keeping secrets from us, especially when it comes to advanced aerospace technology. That's a plausible story. It doesn't take much to say, yeah. I don't know what it is, but here's what I think it is. Aliens are a lot more difficult. It requires the existence of advanced civilization. It requires the ability to have interstellar travel. It requires interest in the earth. And if they've seen our social media feeds, I'm not exactly sure what they're so curious about and why they'd want a closer look. It requires a lot. And so that's why while in some cases it's impossible to rule out aliens, that's quite different from ruling in.

aliens. These are different standards of evidence and all and as scientists, you have to hold ourselves to incredibly high evidentiary standards. Otherwise, we wouldn't have science. So while a personal account or interest or meeting might be very compelling as a human story as it were as a human encounter, as scientists were trained to put away our feelings and approach not just the universe as we have it, but the data as we have it. So where does that leave us? Welcome to science every single day. If you're in a data poor environment where your observations are rough, poorly

organized, difficult to scrutinize, and you squeezed every bit of available information out of the observation, you still can't eliminate a hypothesis that you're kind of done. In science, you make a choice. You either keep working the problem and find other ways to extract information to continue ruling out or ruling in hypotheses.

where you pick up and leave town and move on to other problems where the data are richer and you can make a more compelling argument. That's the way it goes. Every single day across the world and across every field of science, we make observations that we cannot fully explain. We put up our best guess and that's it. That's why there's not a lot of professional astronomical or scientific interest in UFOs or UIPs. The data are poor, rely mostly on eyewitness accounts or very, very rough low-quality recorded data. And so there's not much more to say. It's not anti-aliens. It's not discounting the meaning that people draw from these experiences. It's as scientists we'd rather work on problems that have a lot more data because we can make a lot more progress. That's what we're trying to do. And that's where I'm going to end the episode. Could UAPs and UFOs be aliens? Sure. They could be.

Am I personally convinced from the available evidence? No. And that's a judgment call. And I know that other people make different judgment calls on the same body of evidence that happens all the time in science too. We disagree about the validity or importance of different kinds and classifications of evidence. And if people want to attach meaning to it and build communities around it, sure, people attach meaning and build communities around all sorts of things. But I want to celebrate. what I want to focus on is not the divisions, not that you're right, I'm wrong, or maybe it's the other way around, you're wrong, I'm right. Not that. To me, if people are looking in the sky with wondering curiosity, sharing in the experience of a universe that has full of mysteries, wondering what it's all about and how we're all connected to it, that's my universe too. And I can share in that.

Thanks to Raji and Ken L for the questions that led to today's episode. Thank you for all those questions. That's askaspaceman.com or the website askaspaceman.com. Keep the reviews going on your favorite podcasting platform. That really helps. And of course, thank you for all of your Patreon

contributions. I'd like to thank my top Patreon contributors this month. And I'll see you next time for more complete knowledge of time and space.